

Vivek Pandey

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Full Stack Developer — MERN Stack — REST APIs — Scalable Systems

Summary: Full Stack Developer experienced in building scalable backend systems and high-performance applications. Skilled in MERN stack, REST API development, and AI-integrated backend systems including MCP servers and vector databases.

Education

B.Tech in Computer Science, A.P.J Abdul Kalam Technical University (AKTU)
CGPA: 8.0/10

2022–2026

Experience

Software Developer Intern, Seceon

July 2025 – March 2026

- Built and deployed a Model Context Protocol (MCP) server integrating AI workflows with backend services.
- Integrated Claude AI with MCP server enabling structured tool communication and automation.
- Worked with LanceDB (vector database) and OpenSearch for efficient data indexing and retrieval.
- Deployed and managed 3+ containerized services using Docker, improving scalability and reducing deployment time.

Projects

Deep Packet Inspection Engine

Python — PCAP — TLS — Network Protocols
Code: GitHub

- Built a high-performance DPI system for network traffic analysis and security enforcement.
- Parsed Ethernet, IPv4, TCP, UDP, and TLS protocols from PCAP data.
- Achieved **99.8% accuracy** in HTTPS classification using SNI extraction.
- Processed **50,000+ packets/sec** with optimized pipeline.

Workspace – Team Collaboration Platform

React — Node.js — Express — MongoDB — JWT — REST APIs
Live: Demo — Code: GitHub

- Developed a full-stack collaboration platform with authentication and role-based access control.
- Designed scalable REST APIs for workspace and user management.
- Implemented secure JWT-based authentication and backend architecture.
- Improved user experience by optimizing API response handling and frontend state management.

E-Commerce Web Application

React — Tailwind CSS — Zustand — Clerk Auth
Live: Demo — Code: GitHub

- Built a responsive e-commerce application with product listing and cart functionality.
- Managed global state using Zustand and React Hooks.
- Implemented authentication and protected routes.
- Enhanced performance by reducing unnecessary re-renders and optimizing component structure.

Technical Skills

Languages: Java, JavaScript, Python, SQL

Frontend: React.js, Tailwind CSS, HTML, CSS

Backend: Node.js, Express.js, REST APIs

Database: MongoDB, MySQL, LanceDB

Tools: Docker, Git, Postman, Vercel, Render, OpenSearch

Concepts: Authentication (JWT), API Design, Scalable Systems